# Cognitive Computing
# UCS420

# Recap

- Cognitive Computing and Artificial Intelligence
  - Related but distinct technologies



**Question - Map the following feature of particular technology with their boxes:**

A. **Built to 'think' and decide independently**

B. **Simulate more human-like thought processes to inform human decision-making and not replace it.**

C. **Digital assistant that helps achieve a broader goal by informing the overall decision-making process.**

D. **Works with a user to learn their preferences and respond to more context-dependent information.**

E. **Problem-solving tool to quickly analyzing large amounts of data to recognize patterns and decide based on predefined rules**

F. **Designed to think more like people, build on AI capabilities, but are better at understanding complex, unstructured data. They learn from interactions and provide explanations and recommendations**

| AI | Cognitive Computing |
| --- | --- |

# AI

A. built to ‘think’ and decide independently

E. Problem Solving tool to quickly analyzing large amounts of data to recognize patterns and decide based on predefined rules

# Cognitive Computing

B. simulate more human-like thought processes to inform human decision-making and not replace it.

C. digital assistant that helps achieve a broader goal by informing the overall decision-making process.

D. works with a user to learn their preferences and respond to more context-dependent information.

F. designed to think more like people, build on AI capabilities, but are better at understanding complex, unstructured data. They learn from interactions and provide explanations and recommendations.

# AI

# Cognitive Computing

## Automation versus Augmentation

AI systems are great for automating repetitive or difficult tasks.

Cognitive computing is used to enhance and inform human-based decision-making.

## Specific versus General

AI systems are trained on specific datasets and are good at handling problems with specific answers that can be found

- More contextual, drawing from and responding to different types of inputs.
- Good for open-ended problems and challenges

## Speed versus Accuracy

Can provide solutions quickly, but their output can be limited, inaccurate or not entirely reliable.

- Better solutions but not designed to provide finalized results or complete tasks independently

## Specialization versus Adaptability

Limited by the scope of their training data thus, specialized but not flexible

- More adaptable. Designed to pull from a wider range of variable inputs, cognitive computers can respond to dynamic situations better.

Source: http://ibm.com/think/topics/cognitive-computing-vs-ai

| Feature | Artificial Intelligence (AI) Example | Cognitive Computing Example |
|---|---|---|
| **Automation vs Augmentation** | **Self-checkout machine** automatically scans items and calculates the bill without human intervention. | **IBM Watson for Oncology** analyzes patient records and medical literature to recommend treatment options, while the doctor makes the final decision. |
| **Specific vs General** | **Spam email filter** detects spam emails using patterns learned from a specific dataset. | **ChatGPT** understands questions from different domains (education, healthcare, programming, writing) and responds based on context. |
| **Speed vs Accuracy** | **Face unlock on a smartphone** identifies the user in seconds but may occasionally fail under poor lighting. | **Medical decision-support system** spends more time analyzing symptoms, lab reports, and medical history to provide more reliable recommendations. |
| **Specialization vs Adaptability** | **Chess-playing AI (Deep Blue)** is excellent at chess but cannot perform unrelated tasks like answering questions or diagnosing diseases. | **Virtual healthcare assistant** adapts its recommendations based on the patient's age, medical history, medications, symptoms, and previous interactions. |

# AI vs Cognitive Computing: Difference & Overlap

## ARTIFICIAL INTELLIGENCE (AI)

**PRIMARY GOAL**
Automate tasks and replace human effort.

**APPROACH**
Learns patterns from data and makes autonomous decisions.

**DECISION MAKING**
Makes decisions independently with minimal human involvement.

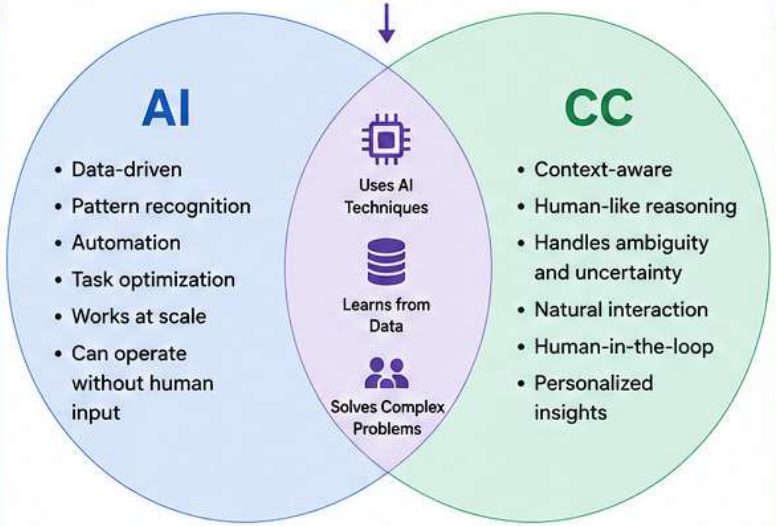
**FOCUS**
Efficiency, automation and scalability.

**OUTPUT**
Predictions, classifications, automation, actions.

**EXAMPLES**
Self-driving cars, facial recognition, spam filters, chatbots, recommendation systems.

**KEY STRENGTH: Speed, Automation, Accuracy**

## OVERLAP




### OVERLAP (Where AI and CC Converge)

| Use of Machine Learning, NLP, Deep Learning and other AI techniques | Extract patterns and knowledge from large amounts of data | Solve complex real-world problems | Improve decision-making and outcomes | Drive innovation across industries |
|---|---|---|---|---|

## COGNITIVE COMPUTING (CC)

**PRIMARY GOAL**
Augment human intelligence and support better decisions.

**APPROACH**
Mimics human reasoning, learning, perception and interaction.

**DECISION MAKING**
Provides recommendations; humans make the final decision.

**FOCUS**
Understanding, reasoning, context and collaboration.

**OUTPUT**
Insights, recommendations, explanations, decision support.

**EXAMPLES**
Clinical decision support, financial advisory, legal research assistants, personalized tutoring.

**KEY STRENGTH: Reasoning, Context, Collaboration**

**AI IN ONE LINE**
AI aims to build systems that can perform intelligent tasks autonomously.

VS

**CC IN ONE LINE**
Cognitive Computing applies AI and insights from cognitive science to create systems that understand, reason and collaborate with humans to improve decision-making.

## KEY DIFFERENCES AT A GLANCE

**Objective**
**AI:** Replace humans
**CC:** Support humans

**Human Involvement**
**AI:** Minimal
**CC:** Active

**Learning Style**
**AI:** Learns patterns
**CC:** Learns with context, experience & interaction

**Problem Solving**
**AI:** Optimizes solution automatically
**CC:** Reasons & explains

**Handling Ambiguity**
**AI:** Struggles with uncertainty
**CC:** Designed for uncertainty

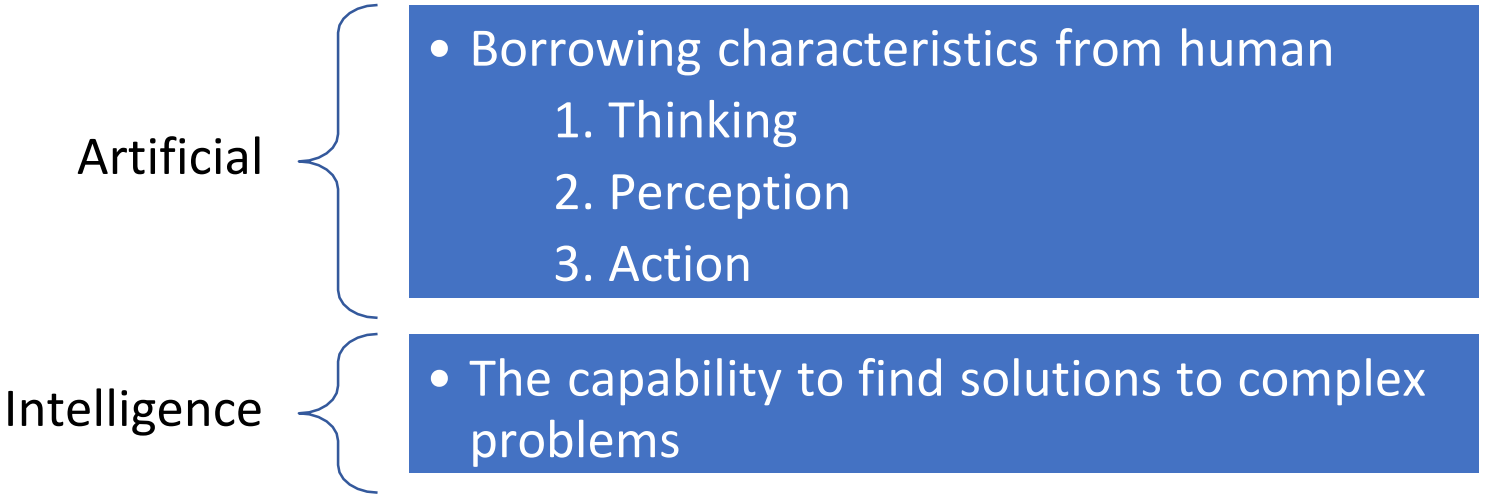
**Communication**
**AI:** Gives answers
**CC:** Gives answers with reasoning

# Lecture - 2

## Cognitive Computing, Cognition, AI, NLP, ML, Big Data

# Cognitive Computing - Introduction

- Cognitive Computing is a subset of **artificial intelligence (AI),** that aims to create systems capable of ***mimicking human-like cognitive functions.***



# Cognition

## Definition

- Cognition refers to the mental processes of acquiring knowledge and understanding through thought, experience, and the senses
- It involves activities like perception, learning, memory, reasoning, and problem-solving
- It is a biological and psychological process that happens in the human brain

## Examples of Processes

- Understanding spoken or written language
- Solving problems and making decisions
- Recognizing patterns and interpreting emotions



# Cognition

## Goal

- To describe and understand how humans think and process information

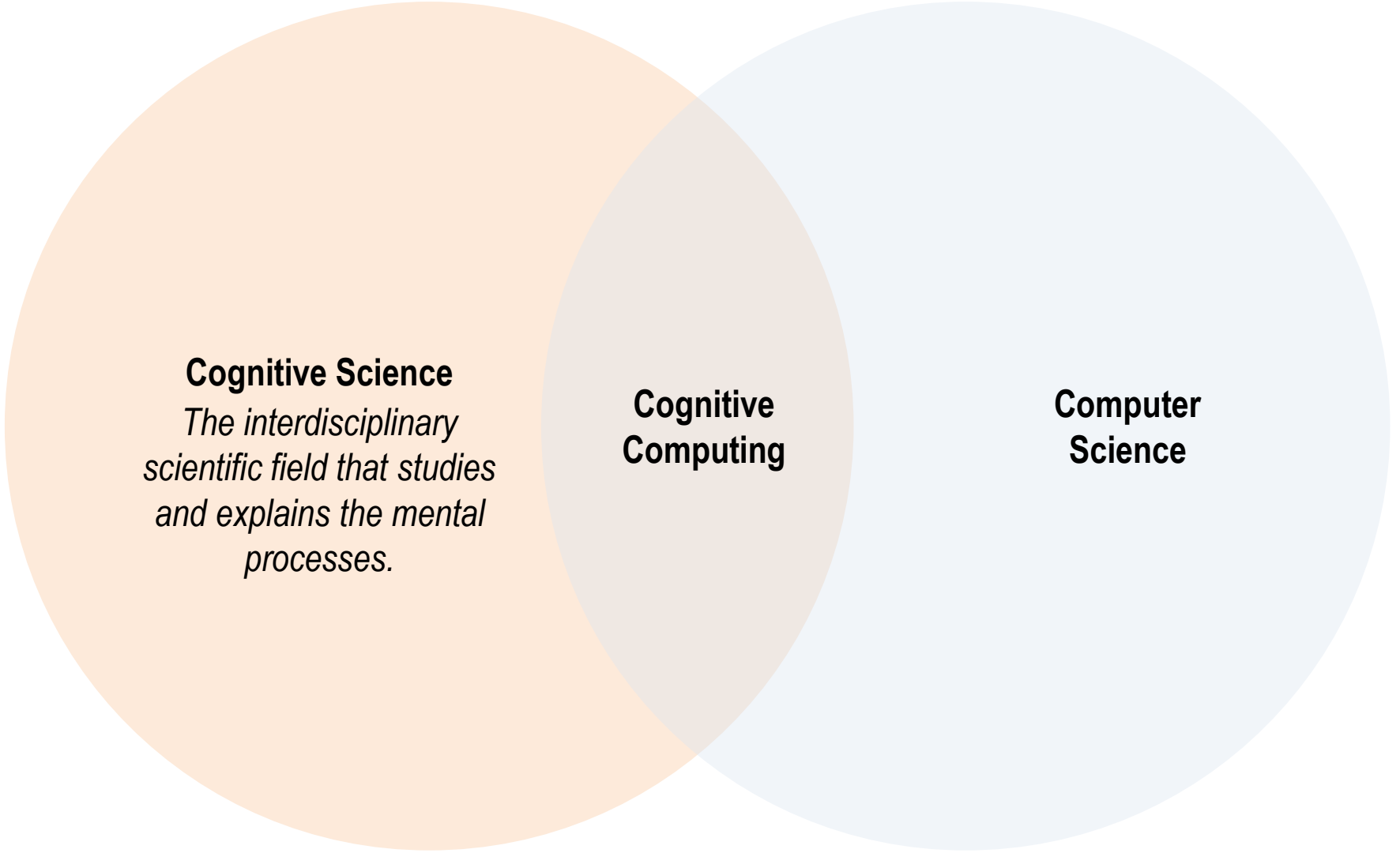
## Key Features

- Involves subjective, emotional, and intuitive elements
- Relates to consciousness and awareness



# Cognition Vs Cognitive Computing

| Aspect | Cognition | Cognitive Computing |
|---|---|---|
| **Definition** | Refers to **the mental processes** of acquiring knowledge and understanding through thought, experience, and the senses. | Refers to **technology systems** that simulate human thought processes.<br>These systems use AI, Machine Learning (ML), Natural Language Processing (NLP), and other tools to mimic human cognition in analyzing data, reasoning, and decision-making. |
| **Focus** | How humans think and understand the world. | Simulating human-like thinking in machines. |
| **Goal** | To study and understand human thought. | To enhance decision-making and problem-solving through machines. |
| **Examples** | Human reasoning, perception, memory. | AI-driven systems like IBM Watson: answering questions in Jeopardy or assisting in medical diagnosis, chatbots. |


Cognitive Science
The interdisciplinary scientific field that studies and explains the mental processes.
Cognitive Computing
Computer Science

# Cognitive Computing - Goal

## Goal

- Develop systems that can understand, interpret, and respond to complex information like the way humans do.
- Focuses on replicating human-like cognitive processes and emphasizes applications where human- machine collaboration is essential.
- AI, on the other hand, encompasses a broader set of technologies and techniques that aim to create intelligent systems capable of performing diverse tasks across various domains.

# Human Intelligence vs AI

## Human Intelligence vs Artificial Intelligence

| Human Intelligence | Artificial Intelligence |
| --- | --- |
| Human intelligence functions on the basis of the surrounding environment. | Artificial intelligence functions on the basis of machine building. It copy's human behavior and create human-like functions. |
| Human brain functions according to memory, emotions, ability to think and computing power. | Artificial intelligence needs specific data and intelligence to function. |
| The ability to make decisions and take rational steps gets slowed down. | Artificial intelligence is a quick process. The ability to give result and take unbiased steps is quicker. |
| Human intelligence can make rational decisions. | Artificial intelligence fails to make rational decisions ending up in taking commands. |
| Human intelligence can have error while making mathematical calculations. | Artificial intelligence, on the other hand, provide very precise calculations. |
| Human brain can perform various functions at a time. They can think, process, and perform various functions at a same time. | Although very quick, AI still lacks behind in the process of multitasking. Every process takes its own time. Even though it is very less, the artificial intelligence does not perform multitasking. |

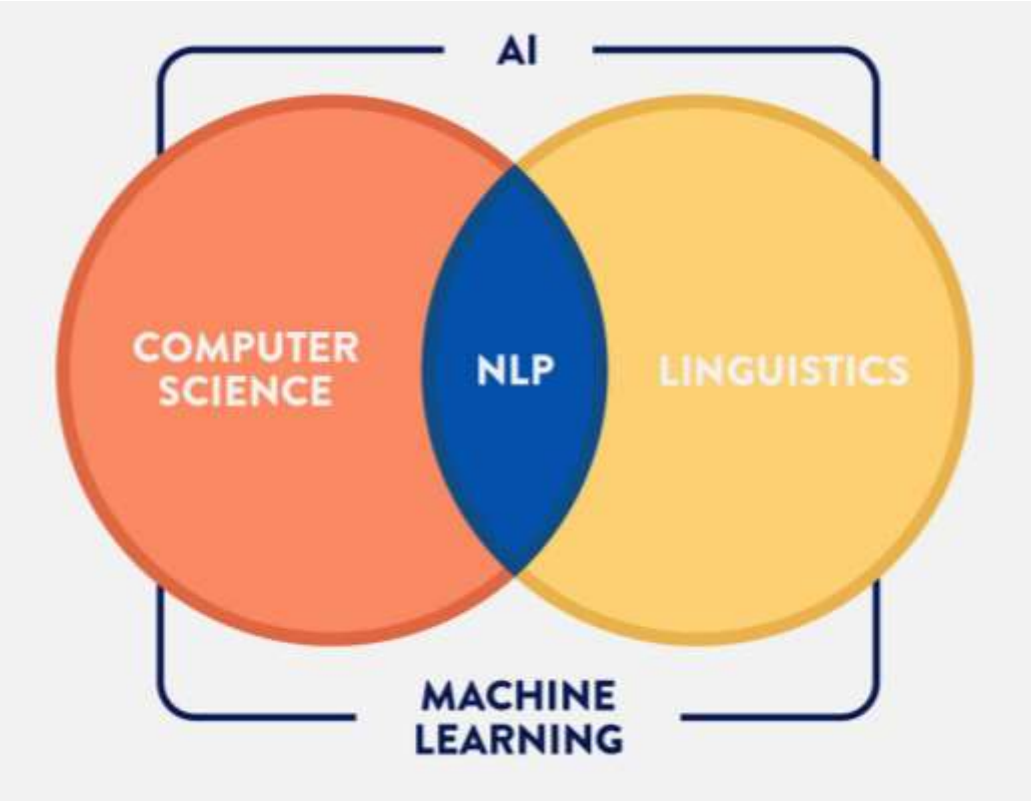


# Foundation of Cognitive Computing

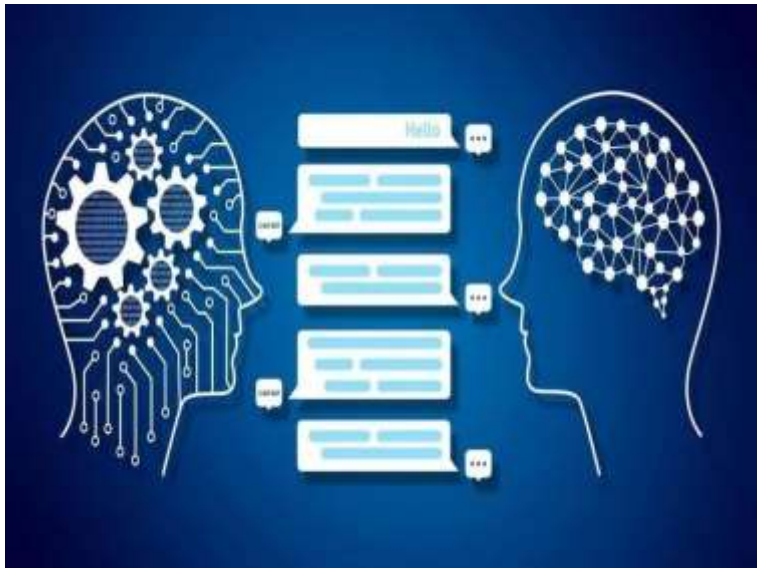
- Natural Language Processing (NLP) (also called Computational Linguistics)
- Machine Learning (ML)
- Big Data Analytics



# Natural Language Processing (NLP)



# Natural Language Processing (NLP)



## Definition

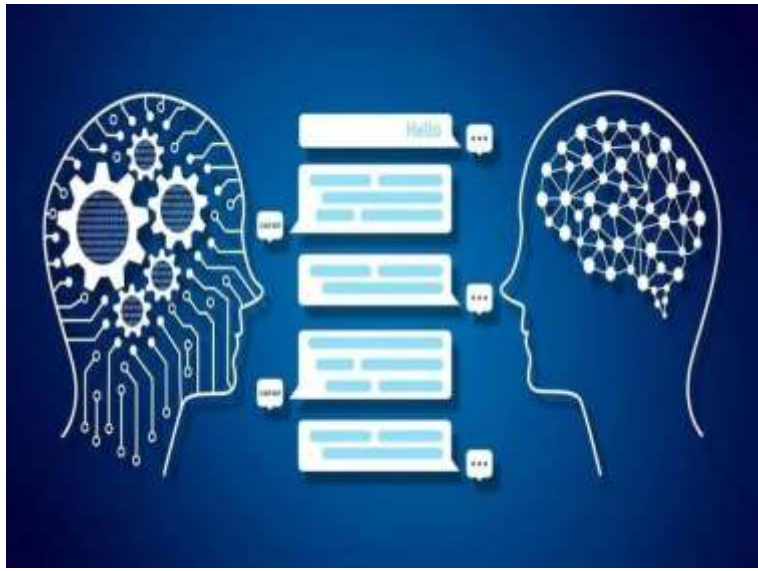
- A field of AI that enables computers to understand and respond to human language

## Key Techniques

- Tokenization
- Parsing
- Sentiment analysis

Also known as **opinion mining**
Used to determine the sentiment or emotion expressed in a piece of text

# Natural Language Processing (NLP)



## Applications

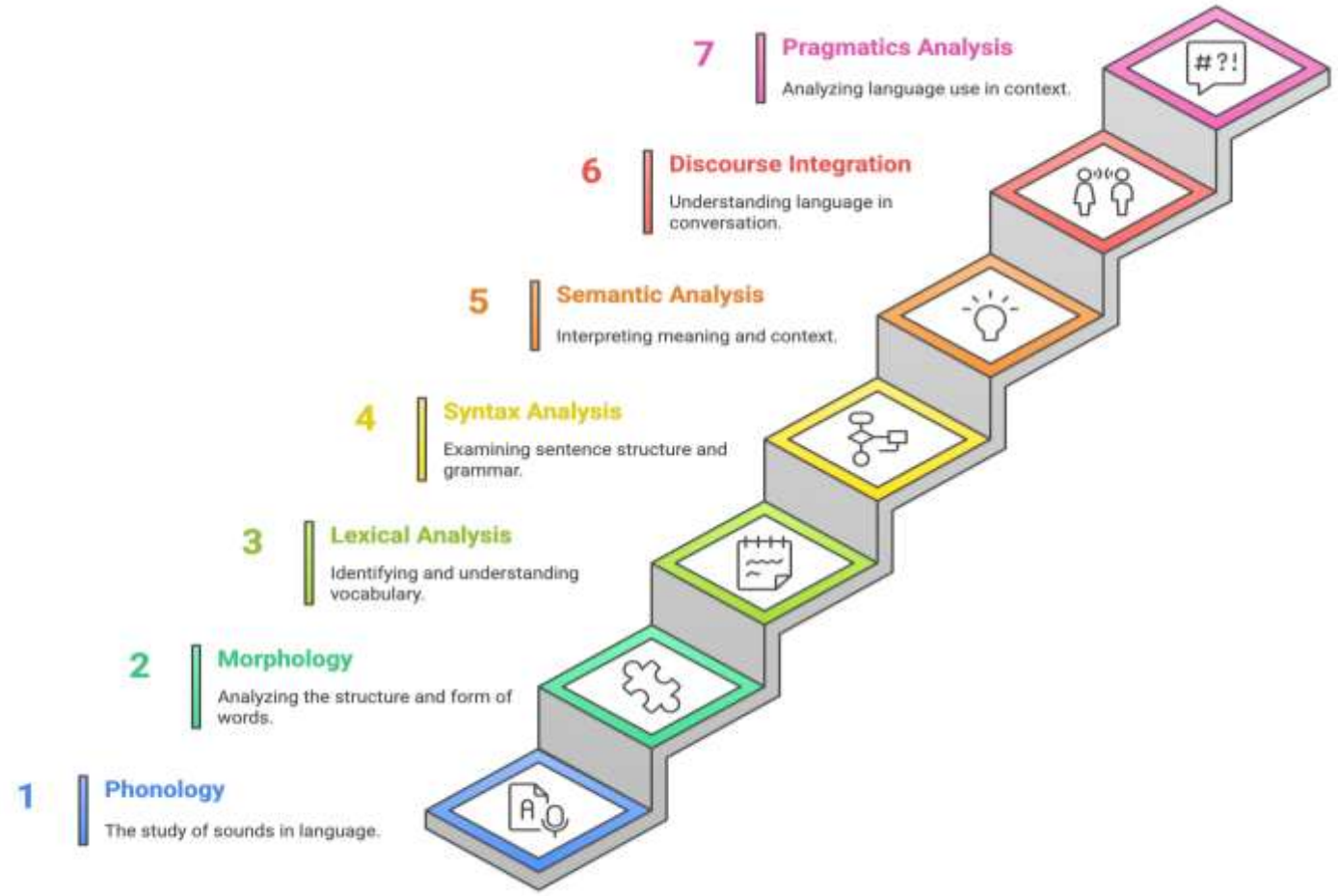
- Chatbots, language translation, smart assistants: Voice assistants like Siri, Alexa.

## Uses in Cognitive Computing

- It allows cognitive systems to interpret human language in all its complexity, enabling machines to understand spoken or written text much like a human listener or reader would.

# Main Steps/Phases/Levels of NLP

1. Phonology
2. Morphology
3. Lexical Analysis
4. Syntax Analysis
5. Semantic Analysis
6. Discourse Integration
7. Pragmatics Analysis



# Natural Language Processing (NLP)

- Examples of sentences
  - ***1. It is the dog that worried the cat, that killed the rat, that ate the malt, that lay in the house, that Jack built***
    - The NLP system must correctly determine:
      - Who worried whom?
      - Who killed whom?
      - What ate the malt?
      - What lay in the house?
      - Simply reading words one by one is not enough.

Dog
└— worried → Cat
└— killed → Rat
└— ate → Malt
└— lay → House
└— built by Jack

  - NLP Challenge
    - This is a syntactic parsing problem.
    - The parser must build a hierarchical parse tree instead of treating the sentence as a sequence of words.

Source: https://www.google.co.in/books/edition/Natural_Language_Processing/FpUBFNFuKWgC?hl=en&gbpv=1&dq=natural+language+processing&printsec=frontcover

# Natural Language Processing (NLP)

- Examples of sentences
  - ***2. I mean what I say vs I say what I mean.***
  - ***3. I see what I eat vs I eat what I see***

Although they contain almost the same words,
they **do not necessarily mean the same thing**.

***These sentences illustrate the complexity of analyzing the natural language***

Source: https://www.google.co.in/books/edition/Natural_Language_Processing/FpUBFNFuKWgC?hl=en&gbpv=1&dq=natural+language+processing&printsec=frontcover

# Natural Language Processing (NLP) - Quiz

- How many legs does a dog have if you call its tail a leg?
- Four.
- Calling a tail a leg doesn't make it one.
- Why is this important?
  - *Language isn't only about words.*
  - *Changing a label does not change reality.*
  - *Humans know this using common sense.*
  - *Machines often struggle.*

Source: https://www.google.co.in/books/edition/Natural_Language_Processing/FpUBFNFuKWgC?hl=en&gbpv=1&dq=natural+language+processing&printsec=frontcover

# Machine Learning (ML)

## Definition

- A subset of AI focused on building systems that learn from data.

## Types

- Supervised
- Unsupervised, and
- Reinforcement Learning

# Machine Learning (ML)

## Key Algorithms

- Decision trees, SVM, k-nearest neighbors, etc.

## Applications:

- Image recognition, Email classification

## Uses in Cognitive Computing:

- Equips systems with the ability to learn from data and experiences, much like humans learn from observation and interaction with the world.
- The more data these systems are exposed to, the more accurately they can make predictions or decisions.

# Big Data Analytics

## Definition

- A field that involves examining large and complex data sets to uncover hidden patterns, correlations, and insights, enabling data-driven decision-making.

## Key Techniques

- Data mining, data visualization, predictive analytics, clustering, and real-time processing.

# Big Data Analytics

## Applications

- Fraud detection, recommendation systems, customer segmentation, healthcare diagnostics, financial risk analysis.

## Uses in Cognitive Computing

- It enables cognitive systems to process and analyze massive volumes of **structured, unstructured and semi-structured data**, extracting meaningful insights to support reasoning, learning, and decision-making processes.

## Structured data

- ***Data that can be processed, stored, and retrieved in a fixed format and structure is fixed/schema-defined.***
- ***It refers to highly organized information that can be readily and seamlessly stored and accessed from a database by simple search engine algorithms.***
- ***Example-The employee table in a company database will be structured as the employee details, their job positions, their salaries, etc., will be present in an organized manner.***

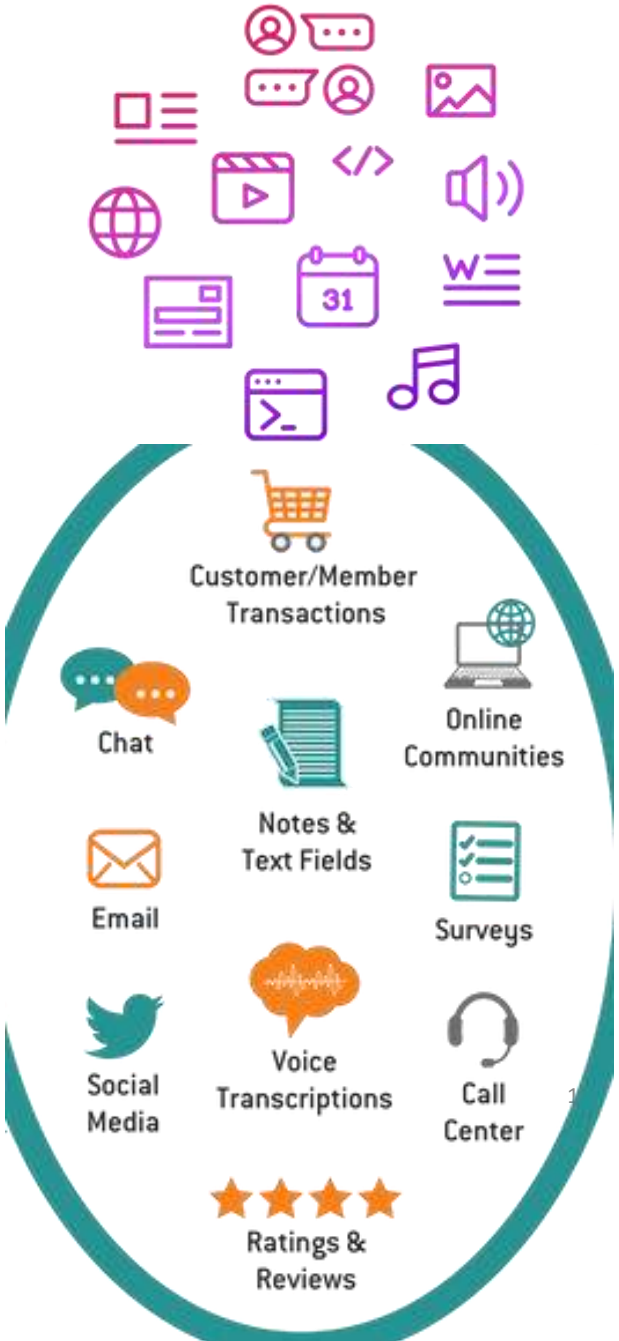
Table1 | Employeeinfo

| ID | Name | F/Name | Position | Salary | Join date | Picture | Phone no | Address |
|---|---|---|---|---|---|---|---|---|
| 1 | Ajmal | Faraz | Manager | $3,000.00 | 1/2/2011 | Package | 435434 | 12, Z westren union bank str |
| 2 | Razaq | Ikbar | Driver | $1,000.00 | 5/2/2011 | Package | 532422 | 5, D avenue street |
| 3 | Moazzam | Nawaz | Security | $1,000.00 | 12/2/2011 | Package | 234235 | 3, D school road |
| 4 | Munir | Muhammad | Doctor | $2,500.00 | 1/4/2012 | Package | 436343 | 5,A national bank |
| 5 | Afzal | Faran | Security | $1,000.00 | 1/8/2012 | Package | 352464 | 9, D park road |
| 6 | Omair | Farhad | Cashier | $2,000.00 | 2/3/2012 | Package | 632345 | 13, Z Chinese restaurant |
| (New) | | | | | | | | |

# Big Data Analytics

## Unstructured data

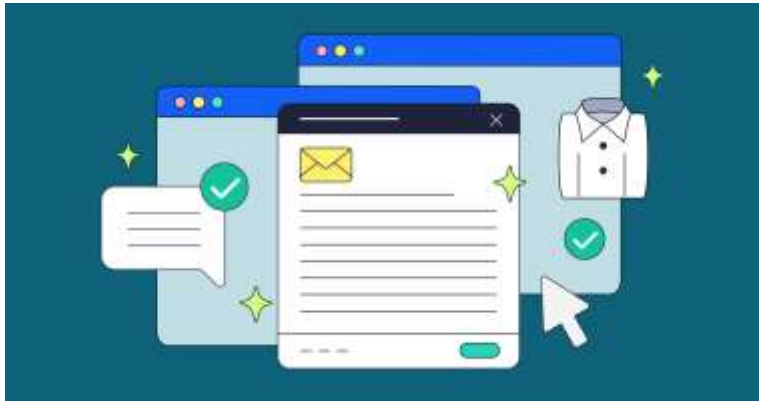
- *Refers to the data that lack any specific form or structure whatsoever.*
- *This makes it very difficult and time-consuming analyze unstructured data.*
- *Stored in data lakes, NoSQL (Not Only Structured Query Language) databases, or cloud-based systems.*
- *Free-form or undefined format (e.g., text, images, videos, audio).*



# Big Data Analytics

## Semi-structured data

- *Pertains to the data containing both the formats mentioned above, that is, structured and unstructured data.*
- *To be precise, it refers to the data that although has not been classified under a particular repository (database), yet contains vital information or tags that segregate individual elements within the data.*
- *Example- Email*
  - *The written content of an email is unstructured, whereas the there are some inherent structure to the information in each email, such as the sender's name, recipient address, recipient name and date sent.*

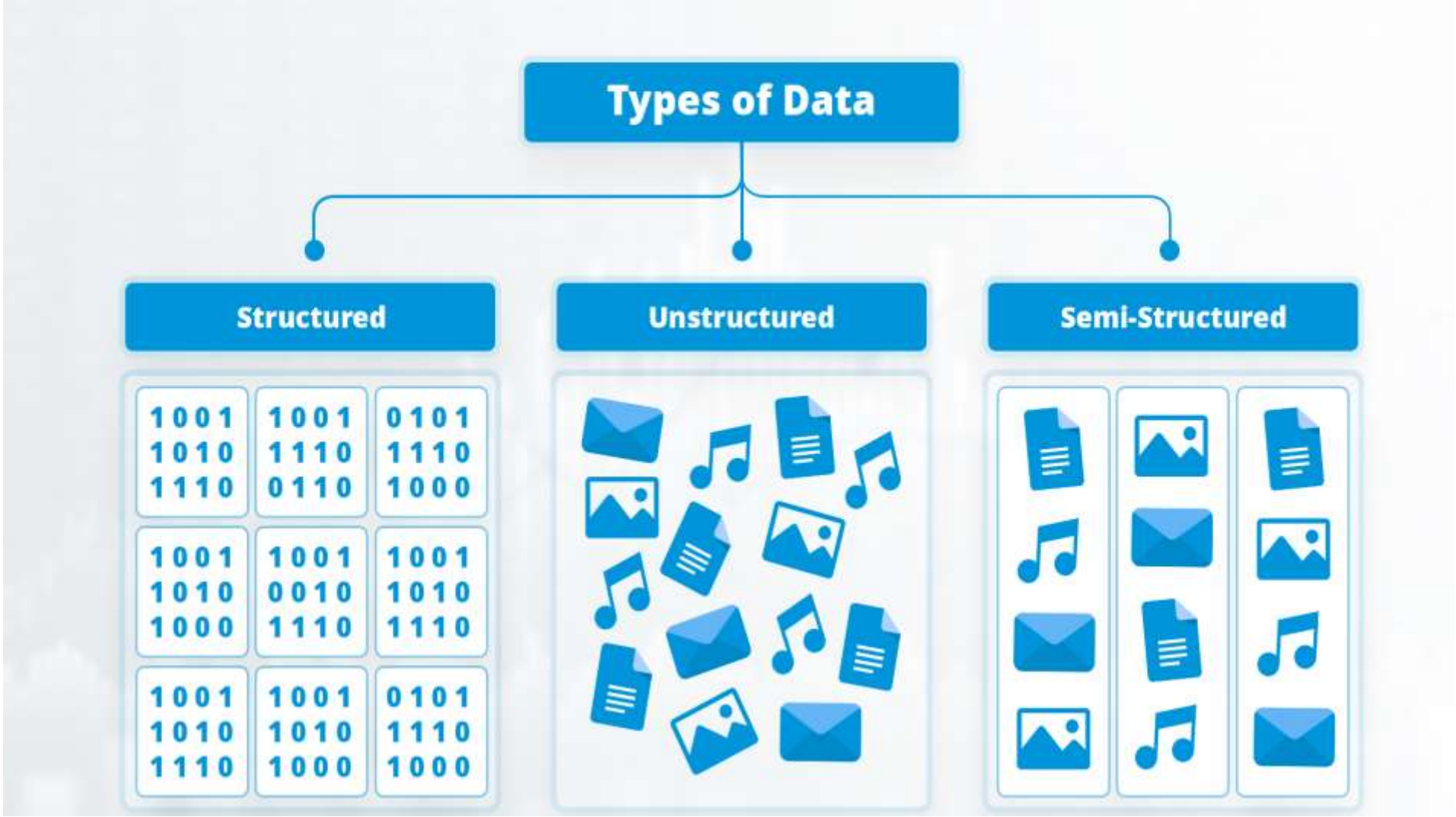


# Example of Semi Structured Data

```
<SocialMediaPost>
  <PostID>12345</PostID>
  <User>
    <Username>TechFan2025</Username>
    <ProfileURL>https://socialmedia.com/TechFan2025</ProfileURL>
  </User>
  <Content>
    <Text>Loving the new features of the iPhone 15! ✨ #iPhone15 #TechLovers</Tex
    <Image>
      <URL>https://socialmedia.com/images/iphone15.png</URL>
      <Description>Image of an iPhone 15</Description>
    </Image>
  </Content>
  <Metadata>
    <Timestamp>2025-01-17T14:35:00</Timestamp>
    <Likes>1200</Likes>
    <Comments>45</Comments>
    <Shares>30</Shares>
  </Metadata>
</SocialMediaPost>
```

# Example of Semi Structured Data

- Uses **XML tags** to organize data.
- The structure is **hierarchical**.
- Different posts may contain different fields (e.g., one post may have a video, another may not).




**Bank Transactions:**
Fields: Transaction ID, Account Number, Date, Amount, Merchant Name.

**Social Media Posts:**
Facebook posts, tweets, Instagram captions, and hashtags.

**Emails with Metadata:**
The body is unstructured, but metadata like sender, receiver, subject, and timestamp is structured.

## QUIZ (True/False):

## 1. CSV spreadsheet is always structured data

## False

*If one column contains free-text (e.g., a customer review or comment field), that column's content is unstructured even though it sits inside a structured table — the file as a whole is better described as semi-structured/mixed.*

## 2. A scanned image of a filled paper form is structured data, since the form has clearly labeled fields (Name, Date, Signature).

## False.

*Visual structure isn't the same as machine-readable structure. Until it's OCR'd/parsed, the scanned image is unstructured data — the computer just sees pixels, not fields.*

**QUIZ (True/False):**

**3. JSON and XML files are unstructured, since they don't use rows and columns like a database table.**

**False**

*JSON/XML have explicit tags/keys defining structure, just not the row-column schema of a relational database — that's exactly why they're the standard example of semi-structured data.*

***Don’t classify data by its surface appearance***

***Classify by what actually matters: whether a fixed, predictable schema exists that a machine can extract without guessing.***

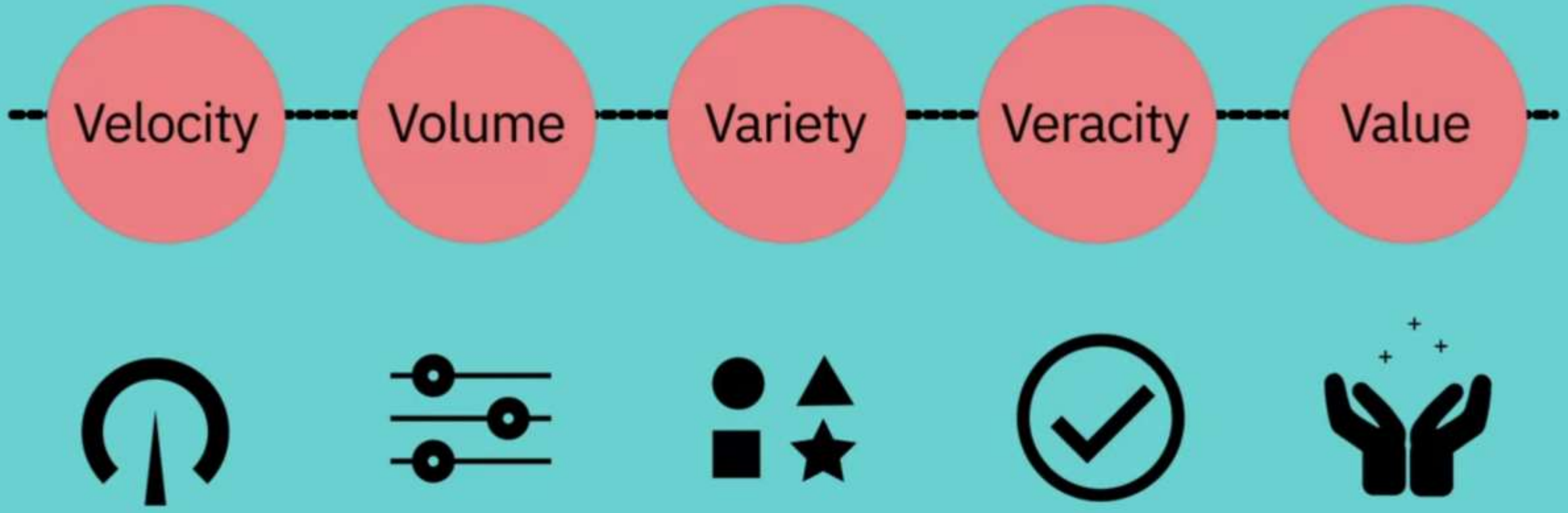
# Big Data

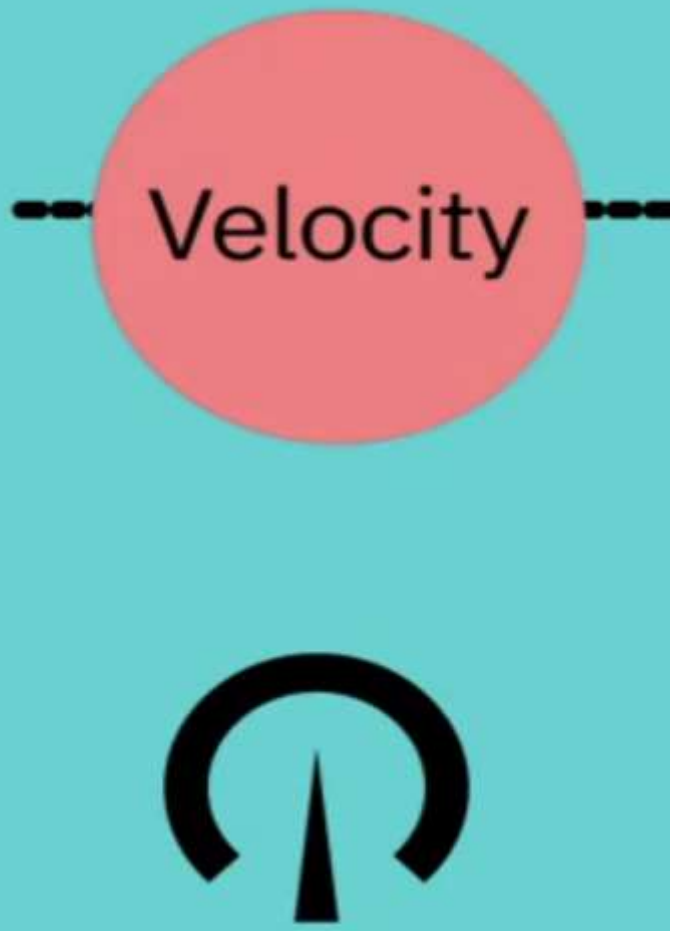
## Definition

- Refers to the **dynamic, large and disparate volumes** of data being created by people, tools, and machines.
- It requires **new, innovative, and scalable technology** to collect, host, and analytically process the vast amount of data gathered in order to derive real-time business insights that relate to consumers, risk, profit, performance, productivity management, and enhanced shareholder value. - Ernst & Young

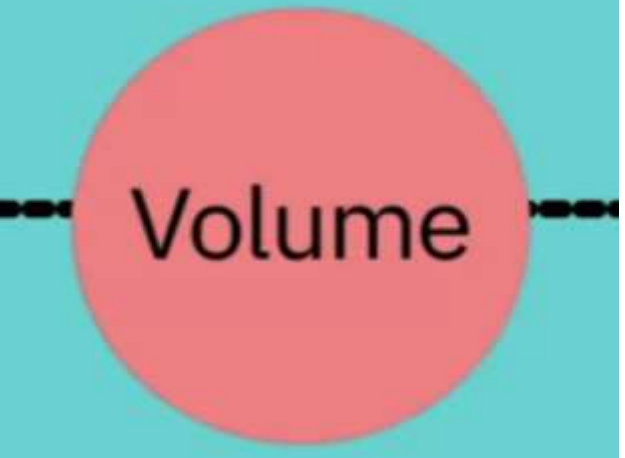
## All definitions have 5 common characteristics

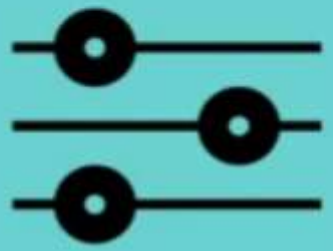
# Characteristics of Big Data – 5Vs – All necessary conditions






- **Velocity is the speed at which data is being created and processed in real time.**
- **Data is being generated extremely fast, in a process that never stops**
- **Near or real-time streaming, local, and cloud-based technologies can process information very quickly**



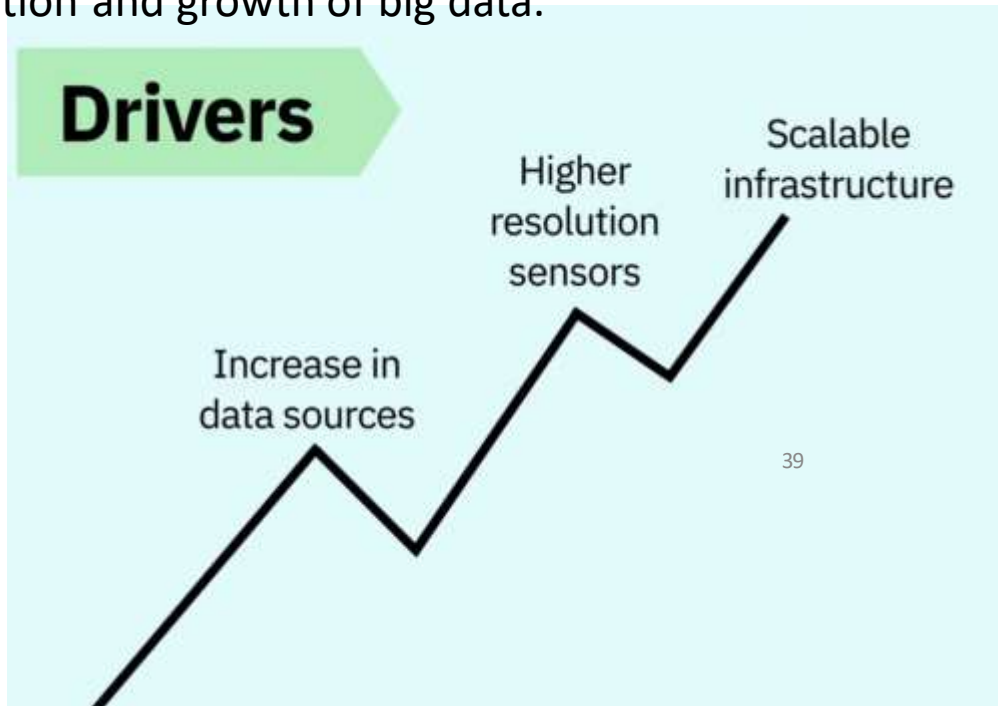



1 Quintillion Bytes = $10^{18}$bytes

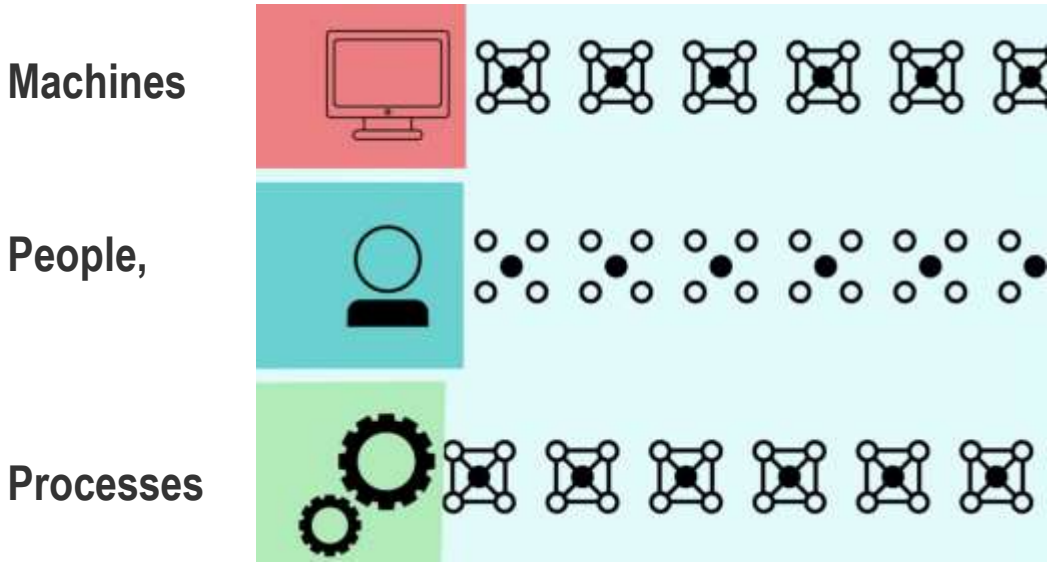
- **Volume is the scale of the data, or the increase in the amount of data stored.**
- **Drivers of volume are the increase in data sources, higher resolution sensors, and scalable infrastructure**
- **A large amount of data is stored in data warehouses.**

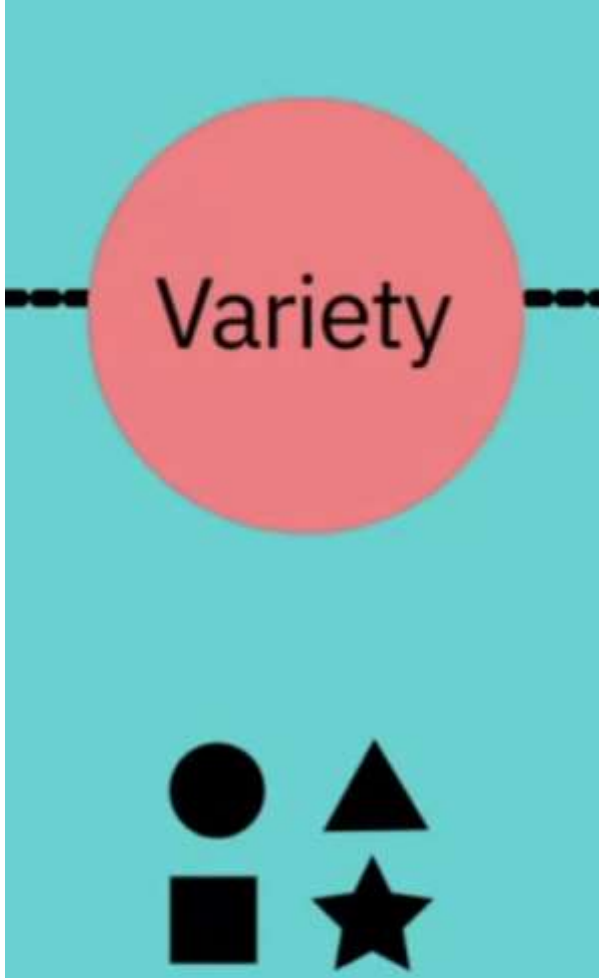
Drivers are the factors that cause the generation and growth of big data.



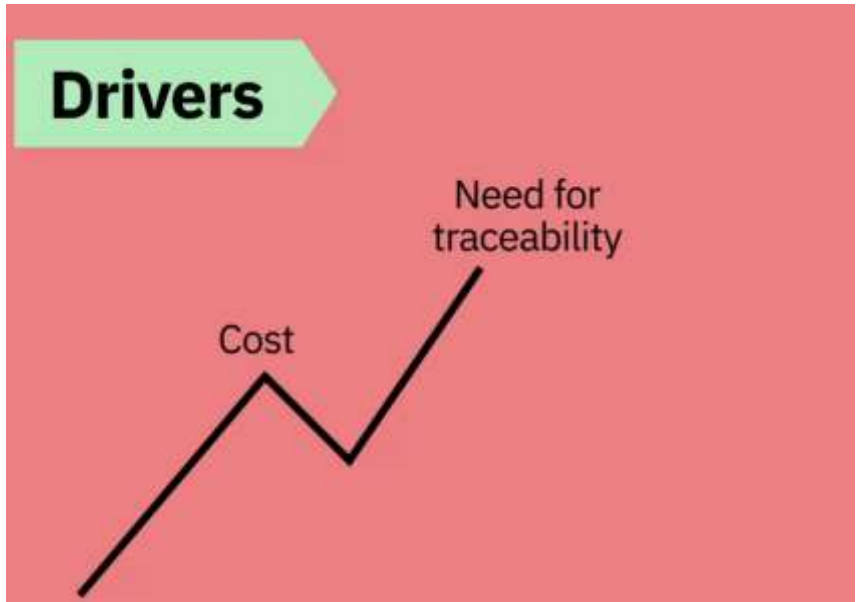
- Variety is the diversity of the data.
- Structured data fits neatly into rows and columns, in relational databases while unstructured data is not organized in a pre-defined way, like Tweets, blog posts, pictures, numbers, and video.
- Variety also reflects that data comes from different sources, machines, people, and processes, both internal and external to organizations.
- Drivers are mobile technologies, social media, wearable technologies, geo technologies, video, and many, many more.

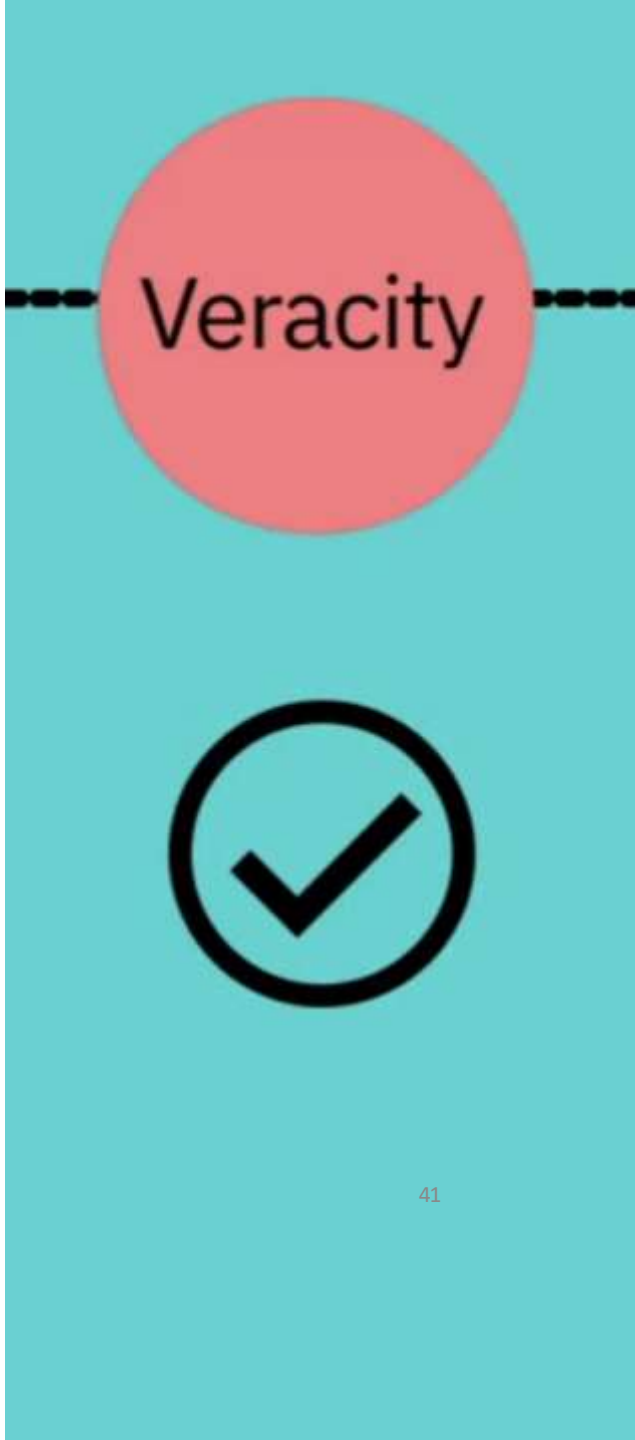


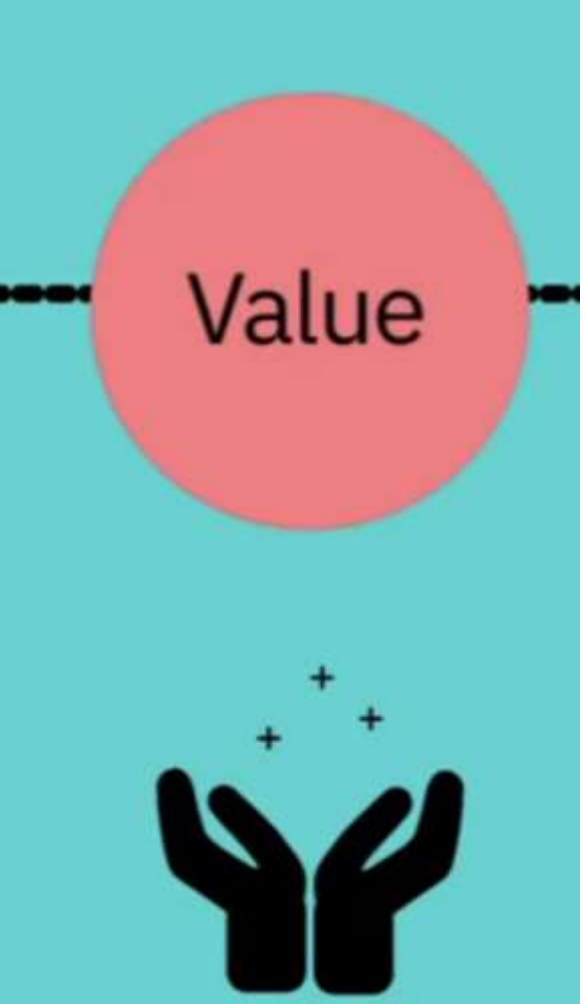



- Veracity is the quality and origin of data, and its conformity to facts and accuracy.
- Attributes include consistency, completeness, integrity, and ambiguity.
- Drivers include cost and the need for traceability.
- With the large amount of data available, the debate rages on about the accuracy of data in the digital age.
- Is the information real, or is it false?



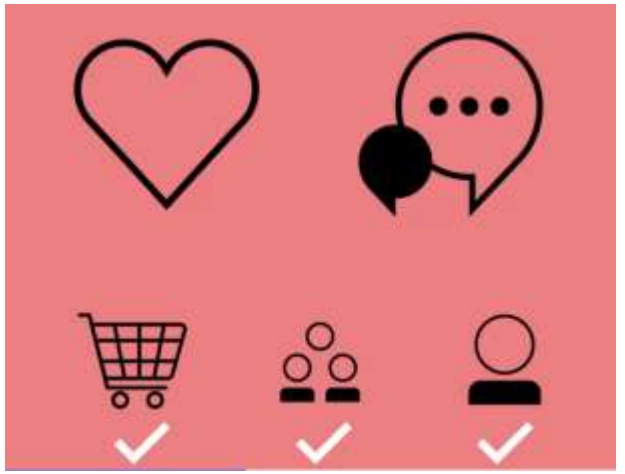






- Value is our ability and need to turn data into value.
- Value isn't just profit.
- It may have medical or social benefits, as well as customer, employee, or personal satisfaction.
- The main reason that people invest time to understand Big Data is to derive value from it.





# Recap Quiz on Big Data

1. **A company has 500 GB of perfectly organized sales data in a single Excel-linked database. Is this Big Data?”**
2. **A hospital collects patient vitals every second from bedside monitors but only reviews the data once a day during rounds. Is this a high-velocity Big Data use case?**
3. **A retail company collects only text data — customer reviews, support tickets, chat logs, and product descriptions. Since it's all text, does this count as low variety?**
4. **A dataset has zero missing values and no typos — every field is perfectly filled in. Does it have high veracity?**
5. **A company has petabytes of Big Data with high volume, velocity, variety, and veracity — but never analyzes it. Does this data have value?**

# Recap Quiz on Big Data - Answers

1. **Not necessarily. Big Data isn't defined by volume in isolation — it's the combination of volume with the other Vs (variety, velocity, etc.) and, crucially, whether traditional tools (a single database/spreadsheet) can still handle it. 500 GB that fits comfortably in a conventional relational database, updated in batches, isn't what the term is pointing at — even though it "sounds like a lot."**

2. **The data generation is high-velocity, but the processing isn't. Velocity refers to both the rate of data creation and the speed required for processing/action — a system can have fast-arriving data but still be architected as batch processing. True high-velocity systems (e.g., ICU alert systems) need near-real-time processing to be useful, not just fast data arrival.**

# Recap Quiz on Big Data - Answers

**3. No — variety isn't about file format; it's about structure and source diversity. Product descriptions (structured/templated), support tickets (semi-structured, with metadata like ticket ID/timestamp), and free-flowing chat logs (unstructured) are three very different data structures despite all being "text." Variety is measured by how differently the data needs to be modeled and processed, not by whether it's technically all the same file type.**

**4. Not necessarily. Veracity is about truthfulness, not just completeness/formatting. A dataset can be perfectly clean and complete while still being biased, manipulated, or simply wrong (e.g., bot-generated fake reviews with flawless formatting, or sensor data that's complete but miscalibrated). Completeness is a necessary check, not a sufficient one for veracity.**

**5. No — Value is not automatic; it must be actively extracted. All four other Vs can be satisfied and the data can still be worthless in practice if no one turns it into a decision, insight, or action. This is the V most often forgotten in exam answers, because students assume "if the data is good, the value is implied" — but Value specifically depends on what's done with the data, not its inherent qualities.**